



MB 552 : Financial Analytics

Course Code : MB 552

Course Name : Financial Analytics

L-T-P-C : 2-0-0-2

Intended for : MBA

Prerequisite :

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction to Financial Analytics:** Overview of Financial Analytics: Importance and Applications; Descriptive, Predictive, and Prescriptive Analytics; Key Financial Data Sources and Data Quality Issues; Role of Statistical Techniques in Financial Analysis. (6 Hours)
- **Data Preparation and Exploration:** Data Cleaning and Transformation Techniques; Financial Data Visualization and Reporting Tools; Exploratory Data Analysis (EDA) Techniques; Identifying Patterns and Outliers in Financial Data. (6 Hours)
- **Predictive Modeling for Financial Analysis:** Regression Models for Forecasting Financial Data; Time-Series Analysis: ARIMA, Seasonal; Decomposition; Machine Learning Techniques: Classification and Clustering; Measuring Model Performance and Avoiding Overfitting. (6 hours)
- **Risk Management and Portfolio Optimization:** Credit Risk Analysis: Credit Scoring Models; Value at Risk (VaR) and Stress Testing; Portfolio Theory and Optimization: Markowitz Model; Hedging and Risk Mitigation Strategies. (6 hours)
- **Applications of Financial Analytics:** Fraud Detection and Prevention Techniques; Algorithmic Trading and High-Frequency Trading; Customer Analytics in Banking and Financial Services; Financial Analytics in Corporate Strategy. (4 hours)

Textbooks:

1. NA

References:

1. NA